

Exercise 3

Prove or disprove the validity of the following sequents:

Part 1: $p \wedge (q \wedge r) \vdash (p \wedge q) \wedge r$

Solution

- | | | |
|----|-------------------------|-----------------|
| 1. | $p \wedge (q \wedge r)$ | premise |
| 2. | p | $\wedge e_1$ 1 |
| 3. | $q \wedge r$ | $\wedge e_2$ 1 |
| 4. | q | $\wedge e_1$ 3 |
| 5. | r | $\wedge e_2$ 3 |
| 6. | $p \wedge q$ | $\wedge i$ 2, 4 |
| 7. | $(p \wedge q) \wedge r$ | $\wedge i$ 5, 6 |

Answer

Valid

Part 2: $\neg p \wedge \neg q \vdash \neg(p \vee q)$

Solution

- | | | |
|-----|------------------------|----------------------|
| 1. | $\neg p \wedge \neg q$ | premise |
| 2. | $\neg p$ | $\wedge e_1$ 1 |
| 3. | $\neg q$ | $\wedge e_2$ 1 |
| 4. | $p \vee q$ | assumption |
| 5. | p | assumption |
| 6. | $\perp$ | $\neg e$ 2, 5 |
| 7. | q | assumption |
| 8. | $\perp$ | $\neg e$ 3, 7 |
| 9. | $\perp$ | $\vee e$ 4, 5–6, 7–8 |
| 10. | $\neg(p \vee q)$ | $\neg i$ 4–9 |

Answer

Valid
